

2.

$$f(x) = 2x^3 - 7x^2 - 10x + 24$$

- (a) Use the factor theorem to show that $(x + 2)$ is a factor of $f(x)$.

(2)

- (b) Factorise $f(x)$ completely.

(4)

3.

$$f(x) = 2x^3 + 5x^2 + 2x + 15$$

(a) Use the factor theorem to show that $(x + 3)$ is a factor of $f(x)$.

(2)

(b) Find the constants a , b and c such that

$$f(x) = (x + 3)(ax^2 + bx + c)$$

(2)

(c) Hence show that $f(x) = 0$ has only one real root.

(2)

(d) Write down the real root of the equation $f(x - 5) = 0$

(1)

4.

$$f(x) = 2x^3 - 5x^2 + ax + 18$$

where a is a constant.

Given that $(x - 3)$ is a factor of $f(x)$,

(a) show that $a = -9$

(2)

(b) factorise $f(x)$ completely.

(4)

Given that

$$g(y) = 2(3^{3y}) - 5(3^{2y}) - 9(3^y) + 18$$

(c) find the values of y that satisfy $g(y) = 0$, giving your answers to 2 decimal places where appropriate.

(3)

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5.

$$f(x) = -6x^3 - 7x^2 + 40x + 21$$

- (a) Use the factor theorem to show that $(x + 3)$ is a factor of $f(x)$

(2)

- (b) Factorise $f(x)$ completely.

(4)

- (c) Hence solve the equation

$$6(2^{3y}) + 7(2^{2y}) = 40(2^y) + 21$$

giving your answer to 2 decimal places.

(3)

6.

$$g(x) = 4x^3 - 12x^2 - 15x + 50$$

(a) Use the factor theorem to show that $(x + 2)$ is a factor of $g(x)$.

(2)

(b) Hence show that $g(x)$ can be written in the form $g(x) = (x + 2)(ax + b)^2$, where a and b are integers to be found.

(4)

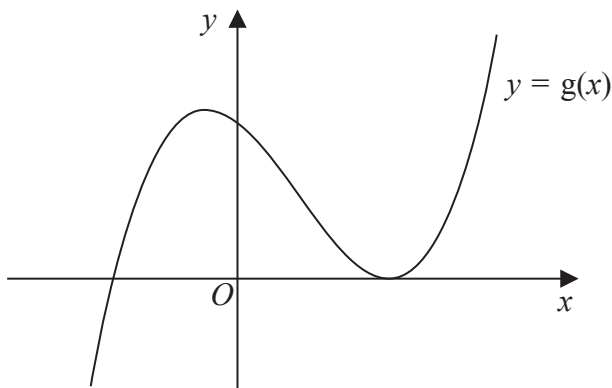


Figure 2

Figure 2 shows a sketch of part of the curve with equation $y = g(x)$

(c) Use your answer to part (b), and the sketch, to deduce the values of x for which

$$(i) \quad g(x) \leq 0$$
$$(ii) \quad g(2x) = 0$$

(3)

7.

$$f(x) = 2x^3 - 13x^2 + 8x + 48$$

(a) Prove that $(x - 4)$ is a factor of $f(x)$.

(2)

(b) Hence, using algebra, show that the equation $f(x) = 0$ has only two distinct roots.

(4)

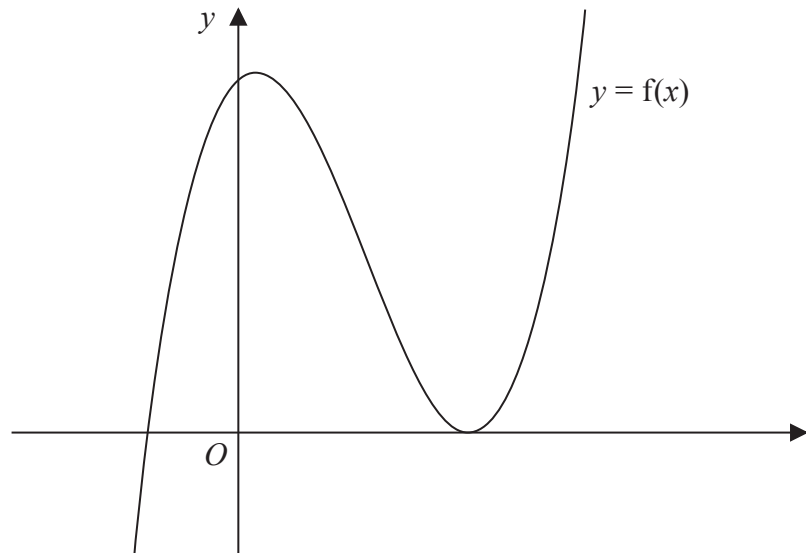


Figure 2

Figure 2 shows a sketch of part of the curve with equation $y = f(x)$.

(c) Deduce, giving reasons for your answer, the number of real roots of the equation

$$2x^3 - 13x^2 + 8x + 46 = 0$$

(2)

Given that k is a constant and the curve with equation $y = f(x + k)$ passes through the origin,

(d) find the two possible values of k .

(2)

8.

$$g(x) = 2x^3 + x^2 - 41x - 70$$

- (a) Use the factor theorem to show that $g(x)$ is divisible by $(x - 5)$. (2)

- (b) Hence, showing all your working, write $g(x)$ as a product of three linear factors. (4)

The finite region R is bounded by the curve with equation $y = g(x)$ and the x -axis, and lies below the x -axis.

- (c) Find, using algebraic integration, the exact value of the area of R . (4)

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9.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

$$f(x) = 4x^3 + 5x^2 - 10x + 4a \quad x \in \mathbb{R}$$

where a is a positive constant.

Given $(x - a)$ is a factor of $f(x)$,

(a) show that

$$a(4a^2 + 5a - 6) = 0 \quad (2)$$

(b) Hence

(i) find the value of a

(ii) use algebra to find the exact solutions of the equation

$$f(x) = 3 \tag{4}$$

10

A curve C has equation $y = f(x)$

Given that

- $f'(x) = 6x^2 + ax - 23$ where a is a constant
- the y intercept of C is -12
- $(x + 4)$ is a factor of $f(x)$

find, in simplest form, $f(x)$

(6)

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11.

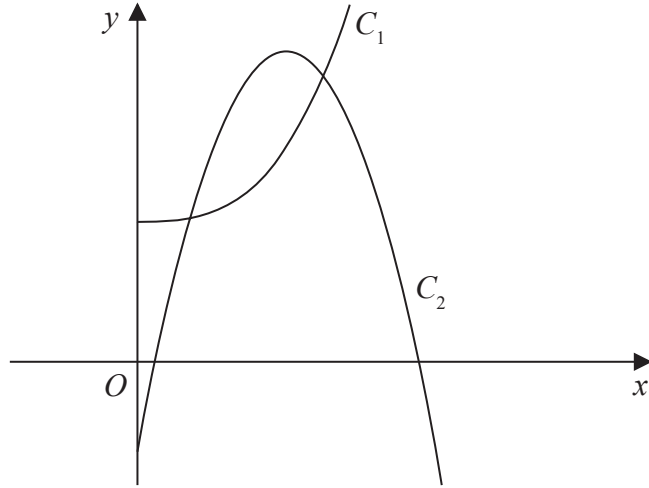


Figure 4

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 2x^3 + 10 \quad x > 0$$

and part of the curve C_2 with equation

$$y = 42x - 15x^2 - 7 \quad x > 0$$

- (a) Verify that the curves intersect at $x = \frac{1}{2}$

(2)

The curves intersect again at the point P

- (b) Using algebra and showing all stages of working, find the exact x coordinate of P

(5)